

Automata and Computability

(Final Assignment Part 1 and Part 2)

Course Code: CSE331

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Problem 1.

Prove the non regularity of the following languages using the pumping lemma.

$$L_1 = \{0^a 1^b 0^c : \text{where } a + b = c\}$$

$$L_2 = \{ww : \text{where, } w \in \Sigma^*\}$$

$$L_3 = \{0^n 1^m : \text{where } n > m\}$$

$$L_4 = \{1^{n^2} : n \geq 0\}$$

$$L_5 = \{1^{2^n} : n \geq 0\}$$

$$L_6 = \{1^n : \text{where, } n \text{ is prime}\}$$

$$L_7 = \{0^n 1^m : \text{where, } n \neq m\}$$

$$L_8 = \{w \in \Sigma^* : \text{where } w \text{ is not a palindrome}\}$$

$$L_9 = \{0^n 1^m : \text{where } n = 4q + r_1, m = 4q + r_2, q \geq 0 \text{ and } 0 \leq r_1, r_2 \leq 3\}$$

$$L_{10} = \{0^n 1^m : \text{where } \frac{n}{m} \text{ is an integer} \}$$

Problem 2 : Design Context Free Grammar(CFG) for the following languages

$$L_1 = \{0^n 1^m : \text{where, } n = 3m\}$$

$$L_2 = \{0^n 1^m : \text{where, } n \leq 3m\}$$

$$L_3 = \{0^i 1^j 2^k : \text{where, } i + j = k\}$$

$$L_4 = \{0^i 1^j 2^k : \text{where, } 2(i + j) = k\}$$

$$L_5 = \{0^i 1^j 2^k : \text{where, } i + j = 2k\}$$

$$L_6 = \{0^i 1^j 2^k : \text{where, } i + k = j\}$$

$$L_7 = \{w : \text{every '0' in } w \text{ is followed by even number of 1's}\}$$

$$L_8 = \{w : w \text{ is a palindrome where no two consecutive symbols are same} \}$$

$$L_9 = \{w : w \text{ has equal number of 0's and 1's}\}$$

$$L_{10} = \{x \# y : x \in \Sigma^*, y \text{ contains exactly two 1's, } |x| = |y|\}$$

Problem 3 :

$$\begin{aligned} S &\rightarrow DQ \\ D &\rightarrow Daa \mid Dab \mid Dba \mid Dbb \mid a \mid b \\ Q &\rightarrow aQb \mid bQa \mid \epsilon \end{aligned}$$

- (a) Give a leftmost derivation for the string 'aabbababa'.
- (b) Draw the parse tree corresponding to the derivation you gave in (a).
- (c) Show that the given grammar is ambiguous by drawing two more parse trees (*apart from the one that you have drawn in (b)*) for the given string in (a).
- (d) Write one five-letter string that has exactly one parse tree in the given grammar.

Problem 4 : Design Push Down Automaton (PDA) for the following languages, $\Sigma = \{0, 1\}$

All the languages in Problem 2 (10 languages)

$$L_{11} = \{w_1 \# w_2 : w_1, w_2 \in \Sigma^*, \text{the number of } 00\text{'s in } w_1 \text{ is equal to the number of } 11\text{'s in } w_2\}$$

$$L_{12} = \{w_1 \# w_2 : w_1, w_2 \in \Sigma^*, w_1 \text{ contains even number of zeros and } |w_1| = |w_2|\}$$

Problem 5 : Design Turing Machine (TM) for the following languages, $\Sigma = \{0, 1\}$

$$L_1 = \{w \in \Sigma^*, w \text{ contains } 01\}$$

$$L_2 = \{w \# w^R : w \in \Sigma^*\} (w^R \text{ means } w \text{ written in reverse order})$$

$$L_3 = \{w_1 \# w_2 : w_1, w_2 \in \Sigma^* \text{ and } \text{count}(w_1, 1) = \text{count}(w_2, 1)\}$$